

# Jannatul Ferdous Srabonee

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[Google Scholar](#) | [LinkedIn](#) | [Portfolio](#)

Merced, California, USA

**Research Interests:** Human-Computer Interaction (HCI), Applied Artificial Intelligence (AI), Accessibility, Brain-Computer Interface (BCI)

## EDUCATION

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### PhD Candidate in Electrical Engineering & Computer Science

**Aug 2023 - Present**

University of California, Merced (UC Merced)

*Merced, CA, USA*

Major: Human-Computer Interactions

CGPA: 3.78/4.0

Advisor: Ahmed Arif

Dissertation: Novel Input and Interaction Techniques for Motor and Vision Impaired Individuals

### BSc in Computer Science & Engineering

**Dec 2015 – Mar 2020**

Khulna University of Engineering & Technology (KUET)

*Khulna, Bangladesh*

Advisor: Muhammad Aminul Haque Akhand

## RESEARCH EXPERIENCE

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### Graduate Student Researcher

**Aug 2023 – Present**

Inclusive Interaction Lab, UC Merced

*Merced, CA, USA*

- Conduct quantitative UX research with diverse populations of blind, low-vision, motor-impaired individuals to understand user behavior, preferences, and technology adoption barriers
- Perform quantitative analysis of user feedback and comparative analysis across different interaction techniques to inform functional design strategies
- Visualize quantitative findings with clear reports and presentations for cross-functional research community

### Research Engineer

**Feb 2022 – Jul 2023**

Advanced Intelligent Multidisciplinary Systems Laboratory

*Dhaka, Bangladesh*

- Conducted research in a multidisciplinary team of neuroscientists, psychiatrists and engineers to develop a depression screening tool using brain-signal data and machine learning
- Developed a machine learning framework to identify depressed groups

### Machine Learning Engineer Intern

**Jul 2021 – Dec 2021**

Robotech Valley

*Dhaka, Bangladesh*

- Analyzed large-scale public dataset of people with and without masks to develop a computer vision based IoT architecture for post COVID-19 prevention measures

### Undergraduate Researcher

**Jan 2019 – Mar 2020**

Intelligence Systems Lab

*Khulna, Bangladesh*

- Developed autism and alcoholism classification system with correlation-based (Pearson's Correlation Coefficient) translation of brain-signals (EEG) and deep learning algorithm convolutional neural network (CNN)

## PUBLICATIONS

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### Conference Papers

- [C.3] BezelBraille: A Braille-Inspired Bezel-Guided Virtual Keyboard for Low Vision and Blind Users  
Isna Alfi Bustoni, **Jannatul Ferdous Srabonee**, Gulnar Rakhmetulla, & Ahmed Arif  
**MobileHCI 2026** [[Paper](#)]
- [C.2] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription  
**Jannatul Ferdous Srabonee**, Ohoud Mosa Alharbi, Wolfgang Stuerzlinger, & Ahmed Arif  
**CHI 2025** [[Paper](#)] [[Presentation](#)]
- [C.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data  
Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique  
**IEEE TENSYP 2020**. [[Paper](#)]

### Journal Articles

- [J.2] Computer Vision-Based IoT Architecture for Post COVID-19 Preventive Measures  
Ahsanul Akib, Kamruddin Nur, Suman Saha, **Jannatul Ferdous Srabonee**, M. F. Mridha  
Journal of Advances in Information Technology (**JAIT**) **2023**. [[Article](#)]
- [J.1] Autism Detection from 2D Transformed EEG Signal using Convolutional Neural Network  
Zahrul Jannat Peya, MAH Akhand, **Jannatul Ferdous Srabonee**, Nazmul Siddique  
Journal of Computer Science (**JCS**) **2022**. [[Article](#)]

### Book Chapter

- [B.1] Alcoholism Detection from 2D Transformed EEG Signal  
Jannatul Ferdous Srabonee, Zahrul Jannat Peya, MAH Akhand, Nazmul Siddique  
International Joint Conference on Advances in Computational Intelligence (**IJCACI**), **Springer**  
**2020**. [[Chapter](#)]

## PROJECTS

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### Eyes-free Text Entry in Smartphones Using a Braille Inspired Virtual Keyboard [Paper](#)

- Improved blind/low-vision users' smartphone text entry speed by 40% and braille proficiency of novices from 12% to 79% across an 8-session study by designing and prototyping a braille-inspired eyes-free keyboard
- Conducted a user study with 12 blind and low-vision participants
- Accepted for publication at MobileHCI 2026

### Depression Screening Tool Using BCI [Code](#)

- Recorded and preprocessed brain-signal data (EEG) using Brain Products' 64-channel EEG device and Brain Vision software
- Built a brain-signal-based (EEG) depression screening framework, achieving 99% classification accuracy by applying data science and fine-tuning machine learning model (SVM)
- Earned national recognition by receiving the Winner Award out of 292 finalist projects at the BASIS National ICT Awards, Bangladesh

## Face Mask Detection to Address COVID-19 Mask-Compliance Gaps [Paper](#) [Code](#)

- Built a low-cost IoT prevention system combining face mask detection, contactless temperature monitoring, and automated disinfectant dispensing, using deep learning (MobileNetV2)
- Achieved high accuracy of 99.88% in detecting masked, improperly masked, and unmasked faces in real-time video streams
- Published in Scopus-indexed journal

## Early Autism Detection from Brain-Signals Using Deep Learning [Paper](#) [Code](#)

- Improved brain-signal-based (EEG) autism detection by applying correlation-based transformation of brain-signal (EEG) and deep learning (CNN)
- Achieved 88% classification accuracy using five-fold cross-validation
- Published in Scopus-indexed journal

## EEG-based Alcoholism Detection [Paper](#)

- Developed an EEG-based alcoholism detection system using deep learning model (CNN)
- Achieved 98.13% while tested on a benchmark EEG dataset
- Published in Springer

## POSTERS & PRESENTATIONS

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### Posters

- [Po.2] AI in Accessibility, Research Week, University of California, Merced. **2026.**
- [Po.1] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy. **2022.** (Virtual)

### Presentations

- [Pr.8] Exploring the Impacts of HEXACO Personality Traits on Text Composition and Transcription, Yokohama, Japan, **CHI 2025.**
- [Pr.7] A Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, Bangladesh Association of Software Information Services National ICT Awards, Dhaka, Bangladesh, **BASIS 2022.**
- [Pr.6] Machine Learning Based Depression Screening Tool from EEG Data to Aid Clinicians, International Conference on 4th Industrial Revolution for the Emerging Future, Dhaka, Bangladesh, **4IR 2022.**
- [Pr.5] EEG-based Major Depressive Disorder (MDD) Identification through Functional Connectivity Analysis, EBRAINS Workshop: Brain Activity across Scales and Species: analysis of Experiments and Simulations, Rome, Italy, **2022.** (Virtual)
- [Pr.4] COVID-19 Defense System, **IEEE YESIST12** Maker Fair, Dhaka, Bangladesh, **2021.**
- [Pr.3] Mobile App for Post COVID-19 Measures, NASA Space Apps Challenge, Dhaka, Bangladesh, **2021.**
- [Pr.2] Alcoholism Detection from 2D Transformed EEG Signal, International Joint Conference on Advances in Computational Intelligence, Daffodil International University, Dhaka, Bangladesh, **IJCACI 2020.**
- [Pr.1] EEG Based Autism Detection Using CNN Through Correlation Based Transformation of Channels' Data, Dhaka, Bangladesh, **IEEE TENSYP2020.**

## TEACHING EXPERIENCE

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### Teaching Assistant, UC Merced

Aug 2023 – May 2026

Algorithm Design and Analysis, Introduction to Human-Computer Interaction, Introduction to Programming, Discrete Mathematics

## CORE SKILLS

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**Languages:** Python, C/C++, MATLAB, JavaScript, HTML, CSS

**AI/ML Frameworks:** TensorFlow, Keras, scikit-learn, OpenCV

**Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly

**Research Methods:** User Studies, Usability Testing, Experimental Design, Mixed-Methods Research, Interviews, Focus Groups, Quantitative Data Analysis, Statistical Hypothesis Testing

## FELLOWSHIPS & AWARD

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- EECS Bobcat Fellowship | \$11,000, (2024, 2025, 2026), University of California, Merced, CA, USA
- Winner Award, 2022, Bangladesh Association of Software Information Services (BASIS) National ICT Awards, Dhaka, Bangladesh
- General Scholarship in Higher Secondary School Certificate examination, 2015, Board of Intermediate and Secondary Education, Dhaka, Bangladesh

## ACADEMIC SERVICES

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### Co-mentor of Undergraduate Interns

Nov 2024 – Present

Inclusive Interaction Lab at UC Merced

Merced, CA, USA

Emeka Sha (CSE), Anvita Kisara (CSE, graduated)

### Reviewer at CHI

Jan 2026 – Present

Articles Reviewed: 1

### Student Volunteer

Apr 2025 - May 2025

CHI 2025

Yokohama, Japan

### Senior Executive Committee Member

Feb 2019 - Feb 2020

IEEE Student Branch KUET

Khulna, Bangladesh

## PROFESSIONAL AFFILIATIONS

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Association for Computing Machinery (ACM)

since Sep 2023

Health Sciences and Research Institute (HSRI) at UC Merced

since Nov 2024

## PROFESSIONAL EXPERIENCE

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### Python Developer

Jun 2022 – Oct 2022

Mir Info Systems Ltd

Dhaka, Bangladesh